HARSH SINGH

AI Engineer | Agentic AI, RAG & Data Engineering (Python, SQL, LangGraph)

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SUMMARY

AI/ML Engineer with hands-on experience building Python and SQL-driven data pipelines and Generative AI systems. Integrated PL/SQL procedures with Python ETL for structured data extraction, and built production RAG pipelines (LangGraph, LangChain, Qdrant) with GDPR-compliant data governance: PII hashing, RBAC, audit logging. Skilled in applying LLMs, embeddings, vector search, and agentic tool-calling workflows to automate data-quality evaluation and reporting, backed by FastAPI/REST API development and Git-based collaboration.

TECHNICAL SKILLS

Data & AI Engineering: Python, SQL (PostgreSQL, MySQL, PL/SQL), ETL Pipelines, RAG Pipeline Design, Chunking Strategies, Data Validation & Curation

Agentic AI & LLMs: LangGraph, LangChain, LlamaIndex, Agent Memory, OpenAI GPT-4o API, Anthropic Claude, Google Gemini, HuggingFace Transformers, Prompt Engineering

Vector & Retrieval Systems: Qdrant, FAISS, Pinecone, Chroma, pgvector, Semantic Retrieval

ML/DL: Scikit-learn, PyTorch, TensorFlow, Model Evaluation, LLM Evals, Bias Mitigation

Backend, Cloud & DevOps: FastAPI, Flask, Node.js, REST APIs, GCP (Compute Engine, Cloud Storage, Vertex AI), Docker, Kubernetes (basics), Git, Rancher

Languages & Databases: Python, C++, PostgreSQL, MySQL, Vector DBs

EXPERIENCE

AI Engineer | ProdGen.ai **Apr 2026 - Jul 2026**

California, USA (Remote)

- Built and maintained production-grade AI/data pipelines integrating LLMs, RAG, and LangGraph-based agent orchestration (LiteLLM, Ollama, Unstructured) with PostgreSQL, deployed via Docker on GCP.
- Improved end-to-end ML pipelines and AI workflows, integrating ML frameworks and APIs into production-ready, tool-calling-capable applications.
- Built automated Evals pipelines to validate LLM output quality, detect failure cases, and enforce data-quality and reliability standards.

Machine Learning Engineer | DevYoddha **Oct 2025 - Mar 2026**

Abu Dhabi, UAE (Remote)

- Integrated PL/SQL procedures with Python ETL pipelines for structured SaaS client data extraction and reporting workflows.
- Built production RAG pipelines (LangChain + Qdrant, 512-token chunks, 64-token overlap) with a GDPR-compliant privacy layer: SHA-256 PII hashing, GCS encryption, per-company RBAC, right-to-erasure cascade.
- Architected a LangGraph multi-agent pipeline with GPT-4o/Claude tool-calling for autonomous candidate assessment (85%+ accuracy); FastAPI + Celery + Redis async backend.
- Engineered an NLP scoring engine (TF-IDF, sentence-transformers, spaCy PhraseMatcher, regex parsing) into a configurable composite ATS score.

Amazon ML Summer School Scholar | Amazon **Jul 2024 - Aug 2024**

- Selected (top national percentile) for Amazon's AI/ML training program; hands-on in model benchmarking, bias mitigation, and RLHF methodologies for production-scale systems.
- Assessed ML model outputs with Amazon engineers; delivered structured feedback on model behavior, optimization strategies, and output quality.

PROJECTS

AI-Powered ATS Resume Screening System **2026**

Agentic AI, LangGraph, RAG, FastAPI, Qdrant, GCP

- Designed a dual-database architecture (PostgreSQL: 9 relational tables; Qdrant: 384-dim embeddings) with a LangGraph agent for autonomous, tool-calling-based screening; zero-shot JD classification via bart-large-mnli (94%+ confidence).
- Built 20+ FastAPI REST endpoints with RBAC, audit logging, and a GDPR erasure flow, backed by a 4-module NLP pipeline (spaCy NER, TF-IDF, sentence-transformers).

Doctor Availability & Appointment Allocation System

Full-Stack, Random Forest, React, Node.js, MySQL

- Built a full-stack app (React/Node.js/MySQL) with a Random Forest model predicting doctor availability at 84% accuracy.
- Optimized the relational schema (8+ tables), reducing query time by 35%.

EDUCATION

B.Tech - Computer Science & Engineering (Artificial Intelligence), CGPA: 6.96 **Aug 2021 - Jun 2025**

Pranveer Singh Institute of Technology, Kanpur, India

Senior Secondary (XII) - CBSE, 71.4% **Apr 2019 - Mar 2020**

Kendriya Vidyalaya No. 1 AFS, Kanpur, India

Secondary (X) - CBSE, 71.8% **Apr 2017 - Mar 2018**

Kendriya Vidyalaya No. 1 AFS, Kanpur, India

ACHIEVEMENTS & CERTIFICATIONS

- Oracle Cloud Infrastructure 2025: Generative AI Professional Certified (1Z0-1127-25)
- Adobe GenSolve Hackathon: Semi-Finalist, Top 5 percentile among 25,000+ participants
- Virtusa Neural Hackathon 2025: Finalist
- LeetCode: 400+ problems solved